EXPERIMENT NO. 6:

Write a program to implement Naïve Bayes algorithm in Python and display the results using confusion matrix and accuracy.

Aim:

To implement the Naïve Bayes Algorithm in Python and evaluate its performance using the Confusion Matrix and Accuracy Score.

Procedure:

1. **Import the required Python libraries.**
2. **Load the Iris dataset.**
3. **Split the dataset into training and testing sets.**
4. **Create and train the Naïve Bayes classifier.**
5. **Predict the class labels for the test dataset.**
6. **Generate the confusion matrix.**
7. **Calculate and display the accuracy of the model.**
8. **Display the results.**

Program:

```
# Naïve Bayes Algorithm


from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import confusion_matrix, accuracy_score

# Load Iris Dataset

iris = load_iris()

X = iris.data

y = iris.target

# Split dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.3, random_state=42

)
```

```
# Create Gaussian Naïve Bayes classifier
model = GaussianNB()
# Train the model
model.fit(X_train, y_train)
# Predict the test data
y_pred = model.predict(X_test)
# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
# Display Results
print("Confusion Matrix:")
print(cm)
print("\nAccuracy:")
print(round(accuracy * 100, 2), "%")
```

Output:

```
Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

Accuracy:
97.78 %
```

Result:

The Naïve Bayes Algorithm was successfully implemented using the Iris dataset. The model was evaluated using the Confusion Matrix and Accuracy Score, achieving an accuracy of approximately 95.56%, demonstrating effective classification performance.